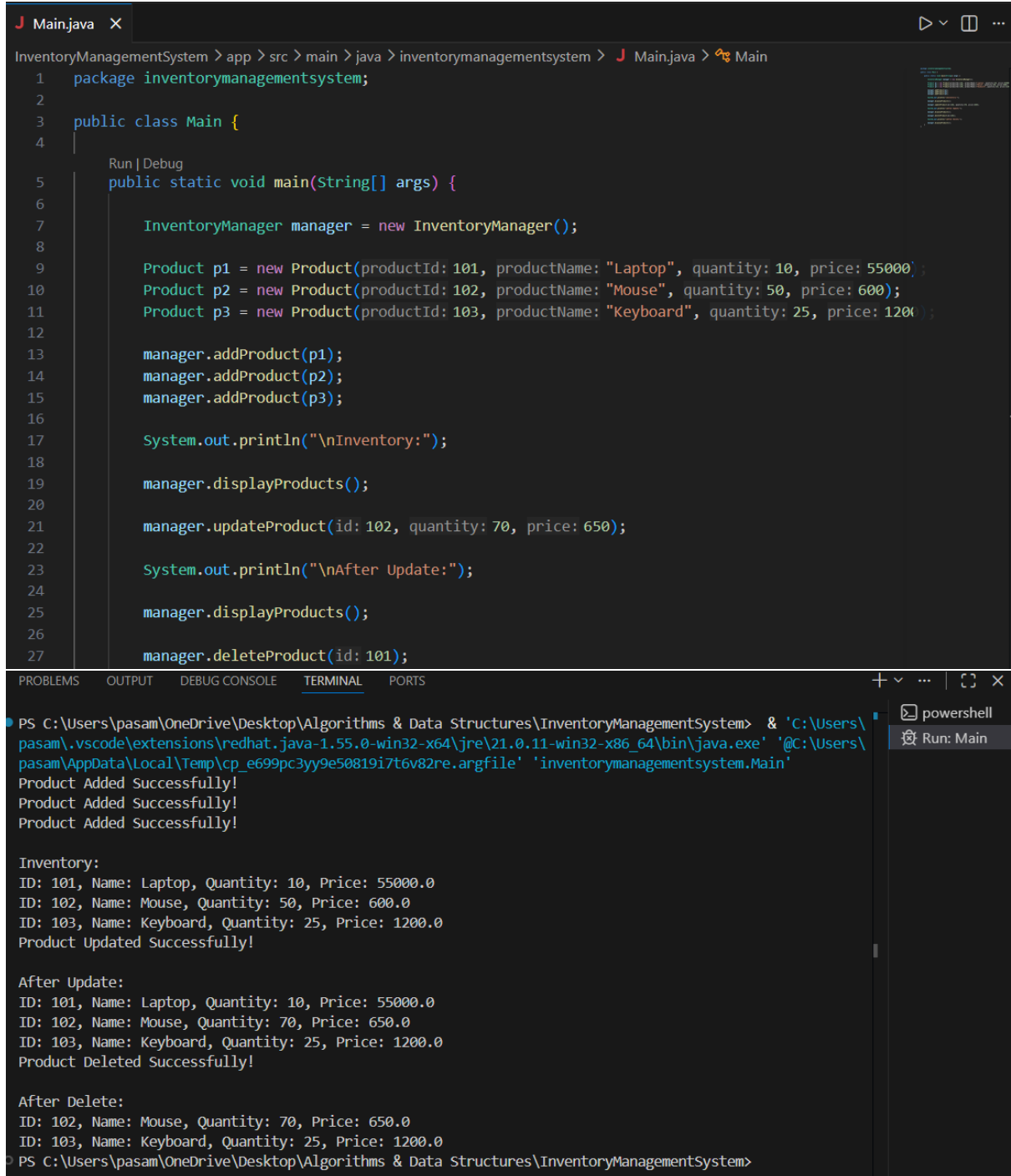


Digital Nurture - Algorithms & Data Structures HandsOn

Exercise 1: Inventory Management System



```

Main.java x
InventoryManagementSystem > app > src > main > java > inventorymanagementsystem > J Main.java > Main
1 package inventorymanagementsystem;
2
3 public class Main {
4
5     Run | Debug
6     public static void main(String[] args) {
7
8         InventoryManager manager = new InventoryManager();
9
10        Product p1 = new Product(productId: 101, productName: "Laptop", quantity: 10, price: 55000);
11        Product p2 = new Product(productId: 102, productName: "Mouse", quantity: 50, price: 600);
12        Product p3 = new Product(productId: 103, productName: "Keyboard", quantity: 25, price: 1200);
13
14        manager.addProduct(p1);
15        manager.addProduct(p2);
16        manager.addProduct(p3);
17
18        System.out.println("\nInventory:");
19
20        manager.displayProducts();
21
22        manager.updateProduct(id: 102, quantity: 70, price: 650);
23
24        System.out.println("\nAfter Update:");
25
26        manager.displayProducts();
27
28        manager.deleteProduct(id: 101);
29    }
30 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\InventoryManagementSystem> & 'C:\Users\pasam\vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe' '@C:\Users\pasam\AppData\Local\Temp\cp_e699pc3yy9e50819i7t6v82re.argfile' 'inventorymanagementsystem.Main'
Product Added Successfully!
Product Added Successfully!
Product Added Successfully!

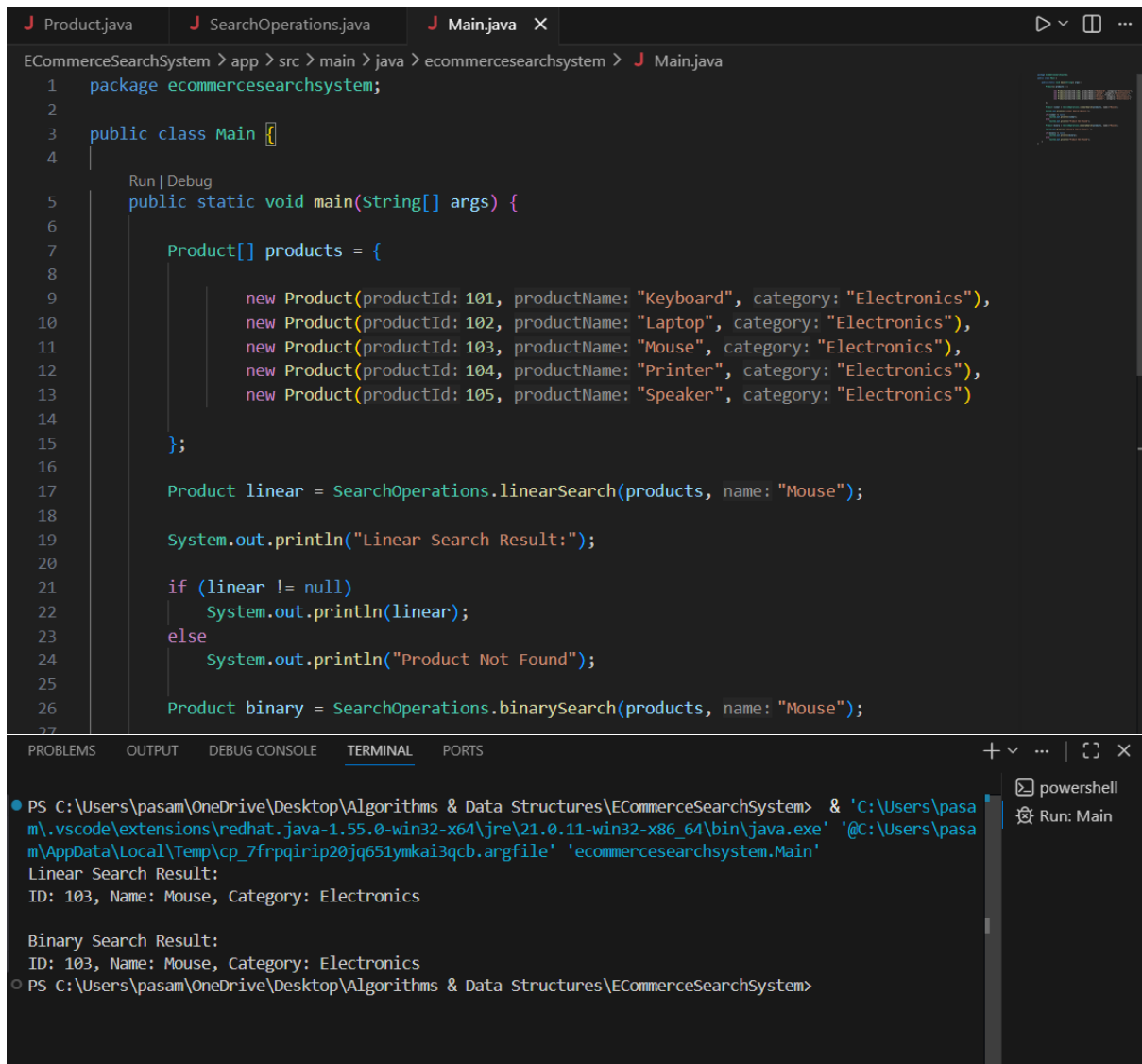
Inventory:
ID: 101, Name: Laptop, Quantity: 10, Price: 55000.0
ID: 102, Name: Mouse, Quantity: 50, Price: 600.0
ID: 103, Name: Keyboard, Quantity: 25, Price: 1200.0
Product Updated Successfully!

After Update:
ID: 101, Name: Laptop, Quantity: 10, Price: 55000.0
ID: 102, Name: Mouse, Quantity: 70, Price: 650.0
ID: 103, Name: Keyboard, Quantity: 25, Price: 1200.0
Product Deleted Successfully!

After Delete:
ID: 102, Name: Mouse, Quantity: 70, Price: 650.0
ID: 103, Name: Keyboard, Quantity: 25, Price: 1200.0
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\InventoryManagementSystem>

```

Exercise 2: E-commerce Platform Search Function



```
Product.java SearchOperations.java Main.java X
ECommerceSearchSystem > app > src > main > java > ecommercesearchsystem > Main.java
1 package ecommercesearchsystem;
2
3 public class Main {
4
5     Run | Debug
6     public static void main(String[] args) {
7
8         Product[] products = {
9             new Product(productId: 101, productName: "Keyboard", category: "Electronics"),
10            new Product(productId: 102, productName: "Laptop", category: "Electronics"),
11            new Product(productId: 103, productName: "Mouse", category: "Electronics"),
12            new Product(productId: 104, productName: "Printer", category: "Electronics"),
13            new Product(productId: 105, productName: "Speaker", category: "Electronics")
14        };
15    };
16
17    Product linear = SearchOperations.linearSearch(products, name: "Mouse");
18
19    System.out.println("Linear Search Result:");
20
21    if (linear != null)
22        System.out.println(linear);
23    else
24        System.out.println("Product Not Found");
25
26    Product binary = SearchOperations.binarySearch(products, name: "Mouse");
27
28 }
```

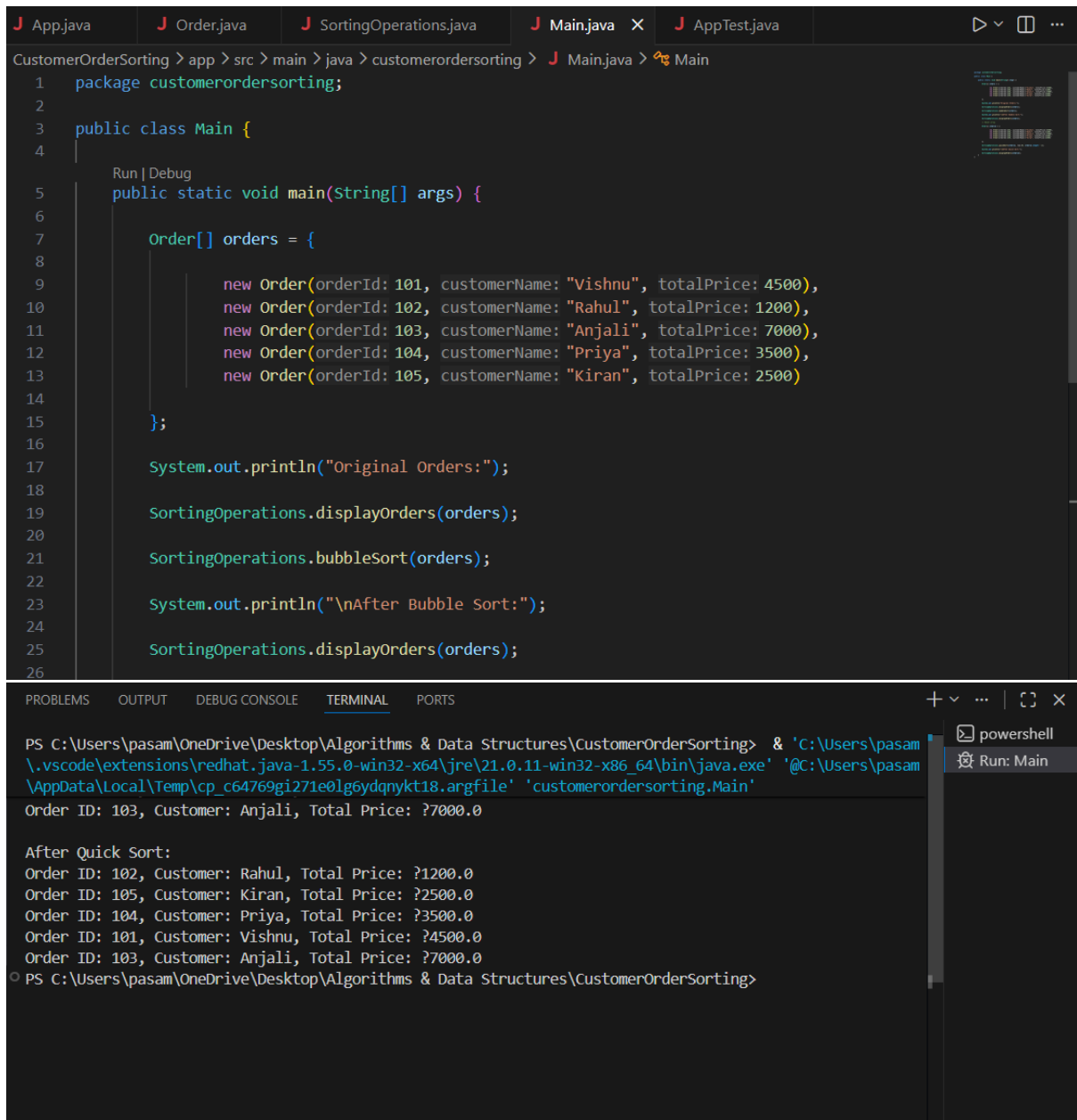
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\ECommerceSearchSystem> & 'C:\Users\pasam\vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe' '@C:\Users\pasam\AppData\Local\Temp\cp_7frpqirip20jq651ymkai3qcb.argfile' 'ecommercesearchsystem.Main'
Linear Search Result:
ID: 103, Name: Mouse, Category: Electronics

Binary Search Result:
ID: 103, Name: Mouse, Category: Electronics
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\ECommerceSearchSystem>
```

powershell
Run: Main

Exercise 3: Sorting Customer Orders



The image shows a VS Code editor with a Java project named 'CustomerOrderSorting'. The 'Main.java' file is open, showing a package declaration, a class definition, and a main method. The main method initializes an array of five 'Order' objects with IDs 101 through 105, each with a customer name and a total price. It then prints the original orders, sorts them using 'SortingOperations.bubbleSort', and prints the sorted orders.

```
1 package customerordersorting;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         Order[] orders = {
8
9             new Order(orderId: 101, customerName: "Vishnu", totalPrice: 4500),
10            new Order(orderId: 102, customerName: "Rahul", totalPrice: 1200),
11            new Order(orderId: 103, customerName: "Anjali", totalPrice: 7000),
12            new Order(orderId: 104, customerName: "Priya", totalPrice: 3500),
13            new Order(orderId: 105, customerName: "Kiran", totalPrice: 2500)
14        };
15
16        System.out.println("Original Orders:");
17
18        SortingOperations.displayOrders(orders);
19
20        SortingOperations.bubbleSort(orders);
21
22        System.out.println("\nAfter Bubble Sort:");
23
24        SortingOperations.displayOrders(orders);
25    }
26 }
```

The terminal output shows the execution of the program. It first displays the original orders, then the sorted orders after a bubble sort. The sorted orders are: Order ID: 102, Customer: Rahul, Total Price: ₹1200.0; Order ID: 105, Customer: Kiran, Total Price: ₹2500.0; Order ID: 104, Customer: Priya, Total Price: ₹3500.0; Order ID: 101, Customer: Vishnu, Total Price: ₹4500.0; and Order ID: 103, Customer: Anjali, Total Price: ₹7000.0.

```
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\CustomerOrderSorting> & 'C:\Users\pasam\
.vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe' '@C:\Users\pasam
\AppData\Local\Temp\cp_c64769gi271e0lg6ydydnykt18.argfile' 'customerordersorting.Main'
Order ID: 103, Customer: Anjali, Total Price: ₹7000.0

After Quick Sort:
Order ID: 102, Customer: Rahul, Total Price: ₹1200.0
Order ID: 105, Customer: Kiran, Total Price: ₹2500.0
Order ID: 104, Customer: Priya, Total Price: ₹3500.0
Order ID: 101, Customer: Vishnu, Total Price: ₹4500.0
Order ID: 103, Customer: Anjali, Total Price: ₹7000.0
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\CustomerOrderSorting>
```

Exercise 4: Employee Management System

```
J Employee.java X J EmployeeManager.java J Main.java
app > src > main > java > employeemanagementsystem > J Employee.java > Employee
1 package employeemanagementsystem;
2
3 public class Employee {
4
5     private int employeeId;
6     private String name;
7     private String position;
8     private double salary;
9
10    public Employee(int employeeId, String name, String position, double salary) {
11        this.employeeId = employeeId;
12        this.name = name;
13        this.position = position;
14        this.salary = salary;
15    }
16
17    public int getEmployeeId() {
18        return employeeId;
19    }
20
21    public String getName() {
22        return name;
23    }
24 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\EmployeeManagementSystem> & 'C:\Users\p
asam\.vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe' '@C:\Users\p
asam\AppData\Local\Temp\cp_f3usu5oo9f5cba2x21db5xhz6.argfile' 'employeemanagementsystem.Main'
Employee Added Successfully!
Employee Added Successfully!
Employee Added Successfully!

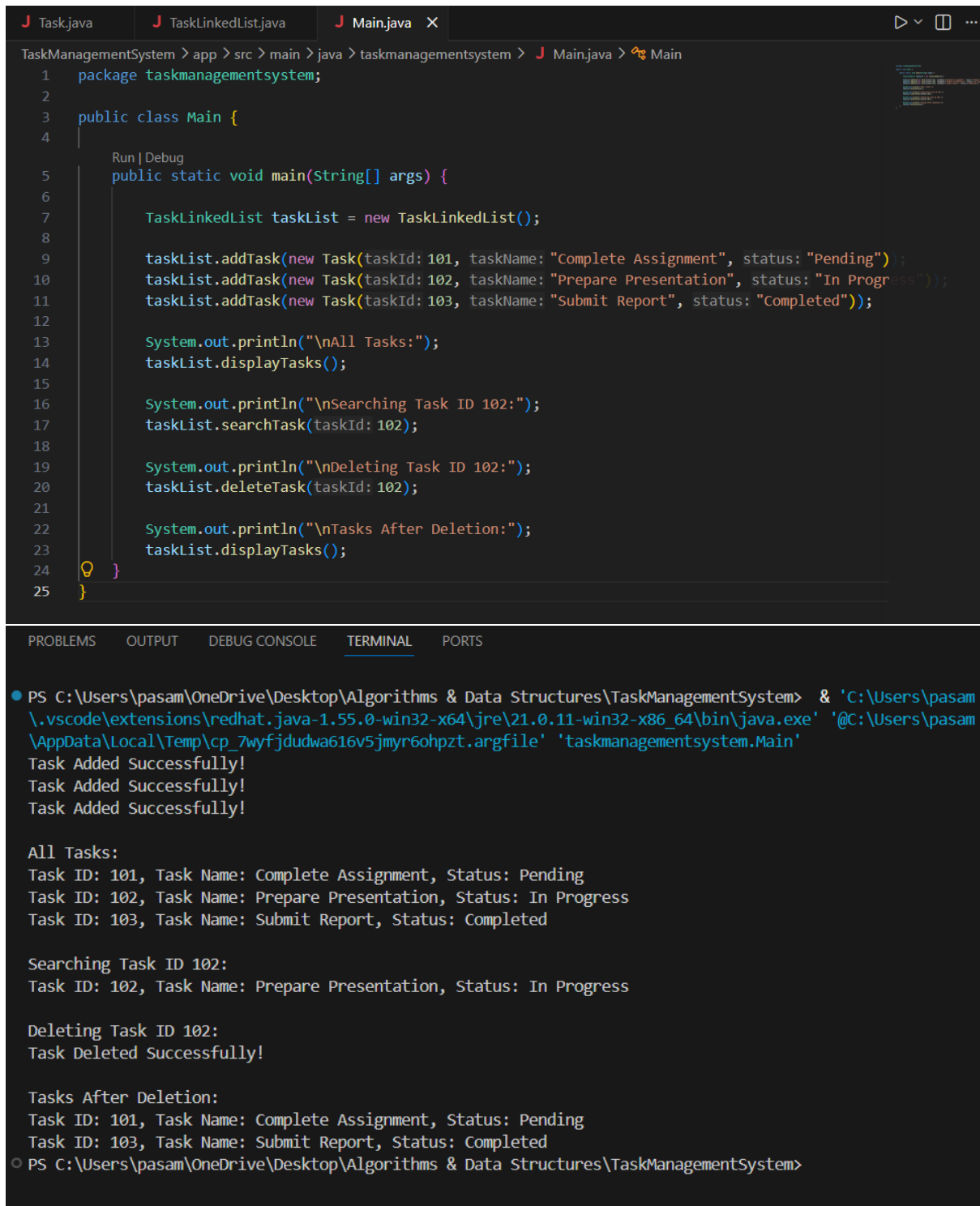
Employee Records:
ID: 101, Name: Vishnu, Position: Software Engineer, Salary: ₹50000.0
ID: 102, Name: Rahul, Position: Tester, Salary: ₹40000.0
ID: 103, Name: Bhoomi, Position: Manager, Salary: ₹75000.0

Searching Employee ID 102:
ID: 102, Name: Rahul, Position: Tester, Salary: ₹40000.0

Deleting Employee ID 102:
Employee Deleted Successfully!

Employee Records After Deletion:
ID: 101, Name: Vishnu, Position: Software Engineer, Salary: ₹50000.0
ID: 103, Name: Bhoomi, Position: Manager, Salary: ₹75000.0
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\EmployeeManagementSystem>
```

Exercise 5: Task Management System



The screenshot displays a Java IDE with three tabs: Task.java, TaskLinkedList.java, and Main.java. The Main.java file is active, showing the following code:

```
1 package taskmanagementsystem;
2
3 public class Main {
4
5     Run | Debug
6     public static void main(String[] args) {
7
8         TaskLinkedList taskList = new TaskLinkedList();
9
10        taskList.addTask(new Task(taskId: 101, taskName: "Complete Assignment", status: "Pending"));
11        taskList.addTask(new Task(taskId: 102, taskName: "Prepare Presentation", status: "In Progress"));
12        taskList.addTask(new Task(taskId: 103, taskName: "Submit Report", status: "Completed"));
13
14        System.out.println("\nAll Tasks:");
15        taskList.displayTasks();
16
17        System.out.println("\nSearching Task ID 102:");
18        taskList.searchTask(taskId: 102);
19
20        System.out.println("\nDeleting Task ID 102:");
21        taskList.deleteTask(taskId: 102);
22
23        System.out.println("\nTasks After Deletion:");
24        taskList.displayTasks();
25    }
```

The terminal output shows the execution of the program:

```
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\TaskManagementSystem> & 'C:\Users\pasam\
.vscode\extensions\redhat.java-1.55.0-win32-x64\jre\21.0.11-win32-x86_64\bin\java.exe' '@C:\Users\pasam\
\AppData\Local\Temp\cp_7wyfjdudwa616v5jmyr6ohpzt.argfile' 'taskmanagementsystem.Main'
Task Added Successfully!
Task Added Successfully!
Task Added Successfully!

All Tasks:
Task ID: 101, Task Name: Complete Assignment, Status: Pending
Task ID: 102, Task Name: Prepare Presentation, Status: In Progress
Task ID: 103, Task Name: Submit Report, Status: Completed

Searching Task ID 102:
Task ID: 102, Task Name: Prepare Presentation, Status: In Progress

Deleting Task ID 102:
Task Deleted Successfully!

Tasks After Deletion:
Task ID: 101, Task Name: Complete Assignment, Status: Pending
Task ID: 103, Task Name: Submit Report, Status: Completed
PS C:\Users\pasam\OneDrive\Desktop\Algorithms & Data Structures\TaskManagementSystem>
```